SkillBridge | Linux Command Line

Navigate and inspect files

Understand paths and practise common file commands in a disposable directory.

The shell runs commands in a current working directory. Run pwd to print that location. Run ls to list visible entries, and ls -la to include hidden names and show detailed information.

An absolute path begins at the filesystem root, such as /home/student/practice. A relative path is interpreted from the current directory. The dot names the current directory; two dots name its parent. A tilde commonly expands to the current user's home directory.

Create a safe practice area with mkdir -p ~/skillbridge-practice, then cd ~/skillbridge-practice. Use printf 'hello\n' > notes.txt to create a small text file, then cat notes.txt to read it. The > operator replaces the destination contents; >> appends instead.

Use cp notes.txt notes-copy.txt to copy the file. Use mv notes-copy.txt archive.txt to rename the copy. Confirm both the current path and the exact target before removing a file with rm. File removal from the shell may not use a recycle bin.

Quotes matter when paths contain spaces. For example, mkdir 'practice folder' creates one directory with a space in its name. Without quotes, the shell separates arguments at spaces.

Checkpoint: explain how pwd and ls help you avoid running a file operation in the wrong directory.

Read permissions and inspect output

Interpret file permissions and combine simple read-only commands.

Run ls -l notes.txt to inspect a file's mode and ownership. The permission groups describe access for the owner, group and others. For regular files, r means read, w means write and x means execute. Directory permissions have different effects: execute permits traversal, while read permits listing names.

The mode rw-r----- gives the owner read/write, the group read-only access and others no access. Use chmod u+x script.sh to add execute permission for the owner of your own practice script. Avoid granting broad permissions just to hide an access error.

A pipe sends standard output from one command to the next command's standard input. For example, ls -l | head -n 5 shows the first five output lines. Use grep 'hello' notes.txt to print matching lines in the practice file.

Use man command-name or command-name --help to check supported options. Read the help before using recursive or force options. Work with a regular user account and request elevated rights only for a task that actually requires them.

Practice: create two text files, list their permissions and search for a word in one file. Explain what changes if you remove write permission from a file you own.

Checkpoint: permissions are an access control mechanism, not a substitute for keeping backups or checking command targets.

Use a disposable practice folder/database. Original instructional material prepared for SkillBridge.